

DD_SIP-05 - Promotional Campaigns - Developer Implementation Guide

Version: 1.0 Status: Draft for review Bundle: 09_catalogs Audience: mid-level backend developers, commercial developers, sales app developers, QA Parent DD: DD_SIP-05 - Promotional Campaigns

1. How To Read This Guide

Promotional campaign is a commercial eligibility and redemption flow.

SIP-05 decides whether a lead/customer/order can receive a campaign offer. If the offer is a discount, SIP-05 calls SIP-03 to create the discount assignment. Billing application happens later in DIS-OP/BIL.

2. Step 1 - Create Campaign Draft

Backoffice creates a campaign:

```
POST /api/promotional-campaigns
Idempotency-Key: campaign-acq-nrb-free-install-july
Content-Type: application/json

{
  "operatorCode": "WIK",
  "campaignCode": "ACQ_NRB_FREE_INSTALL_JULY",
  "displayName": "Free Installation July",
  "campaignType": "ACQUISITION",
  "startDate": "2026-07-01",
  "endDate": "2026-07-31",
  "ownerTeamCode": "COMMERCIAL_ACQ",
  "budgetLimitAmount": 500000,
  "currencyCode": "KES",
  "maxParticipants": 1000
}
```

Insert:

```
insert into promotion_campaign (
  campaign_id,
  operator_code,
  campaign_code,
  display_name,
  campaign_type,
  status,
  start_date,
  end_date,
  owner_team_code,
  budget_limit_amount,
  currency_code,
  max_participants,
  created_by_user_id,
  created_at,
  updated_at
) values (
  'camp-001',
  'WIK',
  'ACQ_NRB_FREE_INSTALL_JULY',
  'Free Installation July',
  'ACQUISITION',
  'DRAFT',
  '2026-07-01',
  '2026-07-31',
  'COMMERCIAL_ACQ',
  500000.00,
  'KES',
  1000,
  'usr-commercial-001',
  now(),
  now()
);
```

3. Step 2 - Add Offer

Attach discount offer:

```
POST /api/promotional-campaigns/camp-001/offers
Content-Type: application/json
```

```
{
  "offerType": "DISCOUNT",
  "discountCode": "ACQ_FREE_INSTALL",
  "assignmentScopeType": "ORDER",
  "validityMode": "ORDER_ONLY",
  "priority": 100
}
```

Validate discount first:

```
GET /api/discounts/ACQ_FREE_INSTALL
```

Insert:

```
insert into promotion_campaign_offer (
  offer_id,
  campaign_id,
  offer_type,
  discount_code,
  bundle_code,
  package_ref,
  assignment_scope_type,
  validity_mode,
  duration_days,
  priority,
  status
) values (
  'pco-001',
  'camp-001',
  'DISCOUNT',
  'ACQ_FREE_INSTALL',
  null,
  null,
  'ORDER',
  'ORDER_ONLY',
  null,
  100,
  'ACTIVE'
);
```

4. Step 3 - Add Target Rules

Example: campaign only for one franchise and one package.

```
POST /api/promotional-campaigns/camp-001/target-rules
Content-Type: application/json
```

```
{
  "rules": [
    {
      "ruleType": "FRANCHISE",
      "operator": "IN",
      "ruleValue": {"franchiseIds": ["fr_01J_NRB_002"]},
      "hardExclusion": true,
      "displayOrder": 10
    },
    {
      "ruleType": "PACKAGE",
      "operator": "IN",
      "ruleValue": {"packageRefs": ["pkg-fiber-100m"]},
      "hardExclusion": true,
      "displayOrder": 20
    }
  ]
}
```

Insert rows:

```
insert into promotion_campaign_target_rule (
  target_rule_id,
  campaign_id,
  rule_type,
  operator,
```

```

rule_value_json,
hard_exclusion,
display_order,
status
) values
('pct-001', 'camp-001', 'FRANCHISE', 'IN', '{"franchiseIds":["fr_01J_NRB_002"]}', true, 10, 'ACTIVE'),
('pct-002', 'camp-001', 'PACKAGE', 'IN', '{"packageRefs":["pkg-fiber-100m"]}', true, 20, 'ACTIVE');

```

5. Step 4 - Add Channel

POST /api/promotional-campaigns/camp-001/channels
Content-Type: application/json

```

{
  "channelCode": "SALES_APP",
  "requiresAgentAttribution": true
}

```

Insert:

```

insert into promotion_campaign_channel (
  channel_id,
  campaign_id,
  channel_code,
  requires_agent_attribution,
  status
) values (
  'pcc-001',
  'camp-001',
  'SALES_APP',
  true,
  'ACTIVE'
);

```

6. Step 5 - Validate Campaign

Run:

POST /api/promotional-campaigns/camp-001/validate

Implementation checks:

1. Dates are valid.
2. At least one active offer exists.
3. Discount/bundle/package refs exist and are active.
4. At least one channel exists.
5. Target rules have valid syntax and known reference types.
6. Budget and participant caps are valid.
7. Approval policy is resolved.

If validation passes:

```

update promotion_campaign
set status = 'READY_FOR_REVIEW',
    updated_at = now()
where campaign_id = 'camp-001'
and status = 'DRAFT';

```

7. Step 6 - Approve And Activate

If approval is required, SIP-05 calls EM-CFG-04. SIP-05 does not own approver steps.

POST /api/approvals/evaluate
Content-Type: application/json

```

{
  "operatorCode": "WIK",
  "moduleCode": "SIP-05",
  "actionCode": "CAMPAIGN_LAUNCH_APPROVAL",

```

```

    "subjectType": "PROMOTIONAL_CAMPAIGN",
    "subjectId": "camp-001",
    "context": {
        "campaignCode": "ACQ_NRB_FREE_INSTALL_JULY",
        "campaignType": "ACQUISITION",
        "budgetLimitAmount": 500000.00,
        "currencyCode": "KES",
        "channelCodes": ["SALES_APP"],
        "hasDiscountOffer": true
    }
}

```

Create local approval reference:

```

insert into promotion_campaign_approval (
    approval_id,
    campaign_id,
    approval_policy_code,
    approval_request_id,
    approval_mode,
    camunda_process_instance_id,
    status,
    assigned_queue_code,
    created_at
) values (
    'pca-001',
    'camp-001',
    'CAMPAIGN_COMMERCIAL_FINANCE',
    null,
    'BPMN',
    null,
    'PENDING',
    'COMMERCIAL_APPROVERS',
    now()
);

```

Start EM-CFG-04 request:

```

POST /api/approvals/requests
Idempotency-Key: approval-campaign-camp-001
Content-Type: application/json

{
    "operatorCode": "WIK",
    "moduleCode": "SIP-05",
    "actionCode": "CAMPAIGN_LAUNCH_APPROVAL",
    "subjectType": "PROMOTIONAL_CAMPAIGN",
    "subjectId": "camp-001",
    "policyId": "apol-campaign-finance-wik-v1",
    "callbackUrl": "/internal/promotional-campaigns/camp-001/approval-outcome",
    "context": {
        "campaignCode": "ACQ_NRB_FREE_INSTALL_JULY",
        "budgetLimitAmount": 500000.00,
        "currencyCode": "KES"
    }
}

```

Patch local reference after EM-CFG-04 returns:

```

update promotion_campaign_approval
set approval_request_id = 'apr-camp-001',
    camunda_process_instance_id = 'cam-proc-camp-001'
where approval_id = 'pca-001';

```

Approver approves through EM-CFG-04 task API. EM-CFG-04 calls SIP-05:

```

POST /internal/promotional-campaigns/camp-001/approval-outcome
Content-Type: application/json

{
    "approvalRequestId": "apr-camp-001",
    "outcome": "APPROVED",
    "decisionComment": "Approved budget and July acquisition terms.",
    "decidedByUserId": "usr-finance-001"
}

```

SIP-05 applies the outcome:

```

update promotion_campaign_approval
set status = 'APPROVED',
    actor_user_id = 'usr-finance-001',
    decision_comment = 'Approved budget and July acquisition terms.',
    decided_at = now()
where campaign_id = 'camp-001'

```

```

    and approval_request_id = 'apr-camp-001'
    and status = 'PENDING';

update promotion_campaign
set status = 'APPROVED',
    updated_at = now()
where campaign_id = 'camp-001';

```

Activate on/after start date:

```

update promotion_campaign
set status = 'ACTIVE',
    updated_at = now()
where campaign_id = 'camp-001'
    and status = 'APPROVED'
    and start_date <= current_date
    and end_date >= current_date;

```

Emit PromotionalCampaignActivated.

8. Step 7 - Eligibility Check From Sales App

Sales app calls:

```

POST /api/promotional-campaigns/eligibility
Content-Type: application/json

```

```

{
  "operatorCode": "WIK",
  "channelCode": "SALES_APP",
  "leadId": "lead-001",
  "orderId": "ORD-1001",
  "customerId": "CUS-100045",
  "packageRef": "pkg-fiber-100m",
  "bundleCode": "FIBER_TV_STARTER",
  "franchiseId": "fr_01J_NRB_002",
  "homepassId": "HP-001",
  "onDate": "2026-07-02"
}

```

Backend finds active campaigns:

```

select c.*
from promotion_campaign c
join promotion_campaign_channel ch on ch.campaign_id = c.campaign_id
where c.operator_code = 'WIK'
    and c.status = 'ACTIVE'
    and c.start_date <= date '2026-07-02'
    and c.end_date >= date '2026-07-02'
    and ch.channel_code = 'SALES_APP'
    and ch.status = 'ACTIVE';

```

Then evaluate target rules:

- FRANCHISE IN ["fr_01J_NRB_002"] passes.
- PACKAGE IN ["pkg-fiber-100m"] passes.
- Duplicate participation check passes.
- Caps check passes.

Response:

```

{
  "eligibleCampaigns": [
    {
      "campaignId": "camp-001",
      "campaignCode": "ACQ_NRB_FREE_INSTALL_JULY",
      "displayName": "Free Installation July",
      "offers": [
        {
          "offerId": "pco-001",
          "offerType": "DISCOUNT",
          "discountCode": "ACQ_FREE_INSTALL",
          "assignmentScopeType": "ORDER"
        }
      ]
    }
  ],
  "rejectedCampaigns": []
}

```

```
}
```

9. Step 8 - Redeem Campaign

When order conversion happens, SALES/FUL calls redeem:

```
POST /api/promotional-campaigns/camp-001/redeem
Idempotency-Key: redeem-camp-001-order-ORD-1001
Content-Type: application/json
```

```
{
  "participantType": "ORDER",
  "participantRefId": "ORD-1001",
  "customerId": "CUS-100045",
  "leadId": "lead-001",
  "orderId": "ORD-1001",
  "franchiseId": "fr_01J_NRB_002",
  "agentUserId": "usr-sales-003",
  "selectedOfferIds": ["pco-001"]
}
```

Insert participation:

```
insert into promotion_campaign_participation (
  participation_id,
  campaign_id,
  operator_code,
  participant_type,
  participant_ref_id,
  customer_id,
  lead_id,
  order_id,
  franchise_id,
  agent_user_id,
  status,
  created_at
) values (
  'pcp-001',
  'camp-001',
  'WIK',
  'ORDER',
  'ORD-1001',
  'CUS-100045',
  'lead-001',
  'ORD-1001',
  'fr_01J_NRB_002',
  'usr-sales-003',
  'SELECTED',
  now()
);
```

10. Step 9 - Create SIP-03 Assignment

For discount offer ACQ_FREE_INSTALL, call SIP-03:

```
POST /api/discount-assignments
Idempotency-Key: camp-001-order-ORD-1001-pco-001
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "discountCode": "ACQ_FREE_INSTALL",
  "scopeType": "ORDER",
  "scopeRefId": "ORD-1001",
  "customerId": "CUS-100045",
  "reasonCode": "CAMPAIGN_ACQUISITION",
  "sourceChannel": "CAMPAIGN",
  "campaignId": "camp-001",
  "franchiseId": "fr_01J_NRB_002",
  "validFrom": "2026-07-02",
  "validTo": "2026-07-31",
  "priority": 100,
  "stackingGroupCode": "INSTALLATION"
}
```

Expected response:

```
{
```

```

    "assignmentId": "da-101",
    "status": "ACTIVE",
    "approvalRequired": false
  }

```

Patch participation:

```

update promotion_campaign_participation
set status = 'REDEEMED',
    assignment_id = 'da-101',
    redeemed_at = now()
where participation_id = 'pcp-001';

```

Important: the discount is now assigned. It is not yet applied to money. DIS-OP/BIL applies it when the order charge/invoice/wallet event is processed.

11. Step 10 - Pause Or End Campaign

Pause:

```

POST /api/promotional-campaigns/camp-001/pause
Content-Type: application/json

{
  "reasonCode": "BUDGET_REVIEW"
}

```

Patch:

```

update promotion_campaign
set status = 'PAUSED',
    updated_at = now()
where campaign_id = 'camp-001'
    and status = 'ACTIVE';

```

End:

```

update promotion_campaign
set status = 'ENDED',
    updated_at = now()
where campaign_id = 'camp-001'
    and status in ('ACTIVE', 'PAUSED', 'APPROVED')
    and end_date < current_date;

```

Existing participation remains auditable. New eligibility must exclude PAUSED and ENDED.

12. Drools Rule Samples

Campaign eligibility should be server-side and can use Drools for configurable targeting.

```

package rules.commercial.campaign

rule "Acquisition campaign requires sales app channel and matching franchise"
when
    $ctx : CampaignContextFact(channelCode == "SALES_APP", franchiseAllowed == true, packageAllowed == true)
    $camp : CampaignFact(campaignType == "ACQUISITION", status == "ACTIVE")
    $out : CampaignEligibilityDecision()
then
    $out.addEligibleCampaign($camp.getCampaignCode());
end

rule "Cap reached blocks redemption"
when
    $camp : CampaignFact(capReached == true)
    $out : CampaignRedemptionDecision()
then
    $out.reject("CAMP_CAP_REACHED");
end

rule "Campaign with budget and discount offer requires finance approval"
when
    $camp : CampaignFact(hasDiscountOffer == true, budgetLimitAmount > 0)
    $out : CampaignApprovalDecision()
then
    $out.setApprovalRequired(true);
    $out.setApprovalPolicyCode("CAMPAIGN_COMMERCIAL_FINANCE");
end

```

13. Suggested Worker Commands

```
campaign:activate-due
campaign:end-due
campaign:cap-sync
campaign:eligibility-cache-refresh
campaign:stuck-redemption-retry
```

Command	Purpose
campaign:activate-due	Activates approved campaigns whose start date has arrived.
campaign:end-due	Ends campaigns past end date.
campaign:cap-sync	Recalculates participant/budget counters.
campaign:eligibility-cache-refresh	Refreshes active campaign read model by channel/franchise.
campaign:stuck-redemption-retry	Retries SIP-03 assignment call for selected participation without assignment id.

14. Error Codes

Code	Meaning
CAMP_CODE_DUPLICATE	Campaign code already exists.
CAMP_INVALID_DATE_RANGE	Start/end dates invalid.
CAMP_EMPTY_OFFER	Campaign has no active offer.
CAMP_DISCOUNT_NOT_ACTIVE	Discount offer is inactive/retired.
CAMP_BUNDLE_NOT_ACTIVE	Bundle offer is inactive/unavailable.
CAMP_CHANNEL_NOT_ALLOWED	Channel is not configured for campaign.
CAMP_NOT_ELIGIBLE	Context failed target rules.
CAMP_CAP_REACHED	Participant or budget cap reached.
CAMP_DUPLICATE_PARTICIPATION	Participant already selected/redeemed campaign.
CAMP_ASSIGNMENT_FAILED	SIP-03 assignment call failed.
CAMP_APPROVAL_REQUIRED	Campaign cannot activate before EM-CFG-04 approval.
CAMP_APPROVAL_REQUEST_FAILED	EM-CFG-04 approval request failed.

15. Minimum Implementation Order

1. Create campaign tables and indexes.
2. Implement create/update campaign APIs.
3. Implement offer API with PLM-CFG-04/SIP-04 validation.
4. Implement target rule API.
5. Implement channel API.
6. Implement EM-CFG-04 approval evaluate/request/callback integration.
7. Implement validation/approval/activation lifecycle.
8. Implement Drools-backed eligibility engine.
9. Implement redeem API with idempotency.
10. Implement SIP-03 assignment adapter.
11. Implement activation/end/cap workers.
12. Emit campaign events.
13. Patch Backoffice campaign screens.
14. Patch Sales App campaign eligibility/redeem flow.

15. Add E2E tests.

16. Test Scenarios

16.1 Acquisition Campaign Redeem

- Campaign active for SALES_APP.
- Lead/order matches franchise and package.
- Eligibility returns campaign.
- Redeem creates participation.
- SIP-03 assignment is created.

16.2 Wrong Franchise Rejected

- Campaign targets fr_01J_NRB_002.
- Request has fr_01J_NRB_999.
- Eligibility excludes campaign with reason FRANCHISE_NOT_ELIGIBLE.

16.3 Cap Reached

- maxParticipants = 1000.
- 1000 redeemed rows exist.
- Next redeem returns 409 CAMP_CAP_REACHED.

16.4 Paused Campaign

- Campaign status PAUSED.
- Eligibility does not return it.
- Existing participations remain visible in reports.

16.5 SIP-03 Failure

- Redeem participation created.
- SIP-03 assignment fails transiently.
- Participation remains SELECTED.
- Worker retries using same idempotency key.

17. Developer Checklist

- Campaign eligibility is server-side.
- Sales app does not decide campaign eligibility offline.
- Campaign redeem is idempotent per participant.
- Discount redemption calls SIP-03 assignment API.
- Campaign approval is configured/executed in EM-CFG-04.
- Drools covers configurable eligibility/redemption/approval decisions.
- No invoice/wallet amount is applied in SIP-05.
- Caps and channel restrictions are enforced.
- Events feed reporting and cache refresh.